

PORT ALBERNI, BC, Canada

WMO: 714750

Lat: 49.320N

Lon: 124.930W

Elev: 250

StdP: 14.56

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	19.7	23.5	14.6	11.6	25.0	19.4	14.7	27.0	14.3	37.3	13.6	33.8	2.7	320	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.0	89.6	65.4	85.2	64.1	80.9	62.9	67.2	84.8	65.5	81.5	63.9	78.0	4.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	81.3	69.9	59.8	77.4	68.5	58.5	73.9	67.1	31.7	84.7	30.3	81.8	29.2	78.3	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	10.2	8.7	DB	13.5	95.6	5.9	3.5	9.3	98.2	5.9	100.2	2.6	102.2	-1.7	104.7	(4)
(5)			WB	13.2	69.7	5.6	2.0	9.2	71.2	5.9	72.3	2.7	73.5	-1.4	75.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	35.9	37.6	41.6	46.9	54.3	59.2	65.0	65.1	59.2	49.3	40.7	35.4
	DBStd	11.86	5.58	4.76	4.47	4.92	5.72	5.25	5.07	4.75	4.85	4.83	6.05	5.37
	HDD50	1994	436	348	263	118	22	1	0	0	1	70	282	453
	HDD65	5892	901	767	726	542	332	189	61	58	184	487	728	917
	CDD50	1726	0	0	2	26	156	276	466	470	276	49	4	1
	CDD65	147	0	0	0	0	1	14	62	62	8	0	0	0
	CDH74	2886	0	0	0	16	147	357	1075	1031	258	1	0	1
	CDH80	1076	0	0	0	2	30	130	425	417	72	0	0	0
Wind	WSAvg	2.8	2.6	2.3	2.5	2.9	3.2	3.3	3.5	3.2	2.5	2.4	2.6	2.6
Precipitation	PrecAvg	72.8	12.0	6.9	7.4	4.2	2.0	1.7	1.0	1.9	2.5	7.9	11.9	12.1
	PrecMax	90.4	22.6	18.9	14.4	9.2	5.9	4.1	2.6	8.4	8.9	17.4	26.0	18.9
	PrecMin	47.4	6.1	0.6	1.5	0.5	0.2	0.6	0.2	0.2	0.1	0.0	3.9	5.6
	PrecStd	11.7	4.3	4.4	3.2	2.4	1.4	0.9	0.7	1.7	2.2	4.4	5.9	3.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.1	53.7	65.1	76.2	83.9	90.8	95.3	94.0	87.6	70.2	57.4	52.9
		MCWB	50.2	48.1	51.6	57.4	60.4	64.8	68.4	66.1	64.5	57.4	54.9	51.0
	2%	DB	49.1	50.2	56.7	67.7	78.0	83.5	89.7	89.4	80.9	64.3	54.1	47.7
		MCWB	47.5	45.9	48.1	52.8	59.0	62.8	66.1	65.4	62.7	55.4	52.3	46.2
	5%	DB	45.8	47.5	52.8	61.7	73.0	77.1	84.8	84.8	75.8	60.2	51.8	44.6
		MCWB	44.7	44.5	46.7	50.4	57.4	61.2	64.8	64.4	61.1	54.6	50.3	43.7
	10%	DB	43.0	44.8	49.8	57.3	67.5	71.8	80.0	79.8	71.1	57.2	49.6	42.2
		MCWB	42.3	42.4	45.4	48.6	55.4	58.8	63.3	63.2	59.6	53.6	48.3	41.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.7	49.9	53.0	58.4	62.1	66.6	70.7	68.8	66.8	61.0	55.8	50.4
		MCDB	51.8	51.8	59.5	71.4	78.3	85.9	92.2	88.0	82.5	65.3	56.8	51.5
	2%	WB	47.2	46.9	50.1	54.5	60.0	64.0	67.9	66.8	64.2	57.6	52.7	45.7
		MCDB	48.4	48.9	54.9	64.1	74.5	80.7	85.4	84.7	76.3	61.2	53.9	46.4
	5%	WB	44.1	44.7	47.7	52.0	58.2	61.7	65.8	65.2	62.4	55.6	50.7	43.2
		MCDB	44.9	46.6	51.1	58.5	70.7	75.2	81.6	81.2	72.4	58.8	51.7	43.9
	10%	WB	42.0	42.6	45.8	49.7	56.2	59.6	63.9	63.6	60.6	53.7	48.5	41.2
		MCDB	42.5	44.1	49.0	55.2	66.0	70.3	78.0	77.5	69.2	56.9	49.5	41.6
Mean Daily Temperature Range	5% DB	MDBR	7.7	12.8	15.4	18.9	21.7	20.8	25.0	25.4	22.7	15.3	9.7	7.1
		MCDBR	10.2	16.8	22.0	29.4	33.8	34.0	36.5	37.2	34.0	21.4	11.9	10.5
		MCWBR	9.0	11.9	13.8	16.2	16.0	15.4	14.8	15.0	16.8	13.7	9.8	9.0
	5% WB	MCDBR	9.8	14.0	18.3	24.0	29.2	30.1	32.6	32.5	28.0	18.2	10.7	9.6
		MCWBR	8.7	10.9	12.4	14.0	14.2	13.9	14.0	13.8	14.5	12.4	9.2	8.5
Clear-Sky Solar Irradiance	taub	0.320	0.315	0.334	0.349	0.373	0.363	0.356	0.354	0.341	0.344	0.324	0.318	
	taud	2.459	2.539	2.497	2.450	2.393	2.444	2.504	2.517	2.545	2.494	2.474	2.414	
	Ebn at Noon	233	268	280	285	281	284	285	281	274	251	229	215	
	Edh at Noon	20	23	28	33	36	35	32	30	27	24	19	18	
All-Sky Solar Radiation	RadAvg	267	552	823	1278	1650	1786	1941	1656	1180	639	319	210	
	RadStd	37	85	107	122	169	192	183	138	100	99	57	30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+185	N/A
(47) Regional (56 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page